Project Overview

The goal is to build an **AI-powered personal and business executive assistant** that can communicate, organize, remember, analyze, coordinate, and execute tasks across multiple services.

1. Main Features

A. Personal Executive Assistant

- Task management
- Reminders
- Notifications
- Daily priorities
- Pending-action tracking
- Appointment management
- Follow-up management
- Automated task execution

B. Email & WhatsApp

- Read/manage emails
- Draft professional messages
- Send messages after authorization
- Automated follow-ups
- Conversation history
- Contact context
- Multilingual communication
- User's communication style learning

C. Google Calendar & Meetings

- Create meetings
- Reschedule/cancel
- Calendar invitations
- Availability management
- Meeting buffers
- Scheduling preferences
- Meeting participation
- Meeting notes
- Action-item extraction

- Meeting summaries

D. AI Memory

Assistant long-term memory:

- Personal preferences
- Meeting preferences
- Travel preferences
- Vendor information
- Contact information
- Communication style
- Previous conversations
- Previous decisions
- Workflow history

E. Google Drive & Documents

- Upload files
- Create folders
- Rename files
- Search/retrieve files
- Share files
- Invoice organization
- Document categorization
- Automated document workflows

F. Payment & Invoice

- Payment information
- Invoice requests
- Vendor notifications
- Invoice follow-up
- Invoice verification
- Drive organization
- Payment records
- Monthly summaries
- Accountant reports

G. Vendor Management

- Request quotations
- Check availability
- Collect responses

- Compare quotes
- Follow up
- Track pending responses
- Supplier communication

H. Voice & Phone

- AI phone calls
- Vendor calls
- Information collection
- Call transcription
- Call summaries
- Follow-up calls
- Escalation

Note: Call recording and automated calling will require jurisdiction/provider-specific consent and compliance handling.

I. Travel

- Flight search
- Flight comparison
- Hotel search
- Itinerary management
- Travel preferences
- Points/rewards integration
- Family travel
- Trip planning

J. Financial Analysis

- Expense analysis
- Transaction categorization
- Spending patterns
- Unusual transaction detection
- Period comparison
- Financial reports
- Excel generation

K. Data & Reporting

- Excel
- CSV
- Charts

- Operational reports
- Financial reports
- Uploaded-file analysis
- Automated summaries

L. CRM & Business Operations

- Lead tracking
- CRM
- Follow-ups
- Team coordination
- Project tracking
- Status monitoring
- Escalation
- Operational workflows

M. Proactive AI

Assistant:

- Pending task detection
- Missing information detection
- Next action suggestion
- Follow-up
- Workflow
- User-important escalation
- Long-running task monitoring

2. Recommended Technology Stack

Layer	**Technology**
Frontend	**Next.js / React + TypeScript**
Backend	**Node.js + Express/Fastify + TypeScript**
Database	**PostgreSQL**
ORM	**Prisma**
Cache	**Redis**
Job Queue	**BullMQ**

AI Service	**Python + FastAPI**
AI/Agent Framework	LangGraph / selected AI libraries
Vector Search	**pgvector**
File Storage	S3-compatible storage
Authentication	OAuth 2.0 + Session/JWT
Email	Gmail API
Calendar	Google Calendar API
Drive	Google Drive API
WhatsApp	WhatsApp Business API / approved provider
Voice	Telephony + STT/TTS provider
Deployment	Docker + Coolify/AWS/VPS
Monitoring	Logs + Metrics + Error Tracking

2. Development Phases

Milestone 1 — Frontend Development
→ UI/UX + complete frontend

Milestone 2 — Backend Development
→ API + database + authentication

Milestone 3 — Full-Stack Integration
→ Frontend ↔ Backend

Milestone 4 — AI Development
→ Python AI service + memory + agent foundation

Milestone 5 — API & Automation Integration
→ Google + WhatsApp + Drive + Email + workflows

Milestone 6 — Advanced AI Features
→ proactive assistant + meeting + voice

Milestone 7 — Advanced Business Features
→ travel + finance + CRM + reporting

Milestone 8 — Testing & QA
→ complete testing + security

Milestone 9 — Deployment
→ production infrastructure + CI/CD

Milestone 10 — Finalization & Handover
→ bug fixes + documentation + handover

Consistent development flow:

Frontend → Backend → Full-stack integration → AI → External APIs/Automation → Advanced features → Testing → Deployment → Handover.